



AQ2.3: Activity Questions 3 - Not Graded



This assignment will not be graded and is only for practice.

Level 1:

1 point

Match the matrices in Column A with their inverses in Column B.

☐

a → → iii

☐

a → → ii

☐

b → → iii

☐

b → → i

☐

c → → i

☐

c → → ii

No, the answer is incorrect.

Score: 0

Accepted Answers:

a → → ii

b → → iii

c → → i

(Use the below information to answer questions 2 and 3).

Consider a system of linear equations

$$\begin{aligned} 2x_1 - x_2 &= 3 \\ x_1 - x_3 &= 3 \\ x_2 - x_3 &= 2 \end{aligned}$$

Let the matrix representation of the above system be $Ax = b$, where $A = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, and $b = \begin{bmatrix} 3 \\ 3 \\ 2 \end{bmatrix}$.

1 point

Choose the set of correct options.

☐

$$A^{-1} = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix} A^{-1} = \begin{bmatrix} 2 & 1 & -1 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

☐


$$A^{-1} = \begin{bmatrix} 1 & -11 \\ 1 & -22 \\ 1 & -21 \end{bmatrix} A^{-1} = \begin{bmatrix} 1 & -11 \\ 1 & -22 \\ 1 & -21 \end{bmatrix}$$

☐

$$\text{Adjoint of the matrix } AA \text{ is } \begin{bmatrix} 1 & -11 \\ 1 & -22 \\ 1 & -21 \end{bmatrix} \begin{bmatrix} 1 & -11 \\ 1 & -22 \\ 1 & -21 \end{bmatrix}$$

☐

$$\det(A) = 1 \det(A) = 1.$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$A^{-1} = \begin{bmatrix} 1 & -11 \\ 1 & -22 \\ 1 & -21 \end{bmatrix} A^{-1} = \begin{bmatrix} 1 & -11 \\ 1 & -22 \\ 1 & -21 \end{bmatrix}$$

$$\text{Adjoint of the matrix } AA \text{ is } \begin{bmatrix} 1 & -11 \\ 1 & -22 \\ 1 & -21 \end{bmatrix} \begin{bmatrix} 1 & -11 \\ 1 & -22 \\ 1 & -21 \end{bmatrix}$$

$$\det(A) = 1 \det(A) = 1.$$

1 point

Choose the set of correct options.

☐

$$x_1 = -2x_1 = -2.$$

☐

$$x_2 = 1x_2 = 1.$$

☐

$$x_3 = -1x_3 = -1.$$

☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$x_2 = 1x_2 = 1.$$

$$x_3 = -1x_3 = -1.$$

1 point

Choose the set of correct options.

☐

A system of linear equations $Ax = b$ is called a homogeneous system of linear equations if $b = 0$.

☐

A system of linear equations $Ax = b$ is called a non-homogeneous system of linear equations if $b \neq 0$.

☐

If v is a solution of the system of linear equations $Ax = b$, then cv is a solution of system of linear equations $cAx = b$, where $c \neq 0$.

☐

Let $AX = b$ be a system of linear equations. If A is invertible, then $adj(A)x = b$ also has a solution.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A system of linear equations $AX = b$ is called a non-homogeneous system of linear equations if $b \neq 0$.

If v is a solution of the system of linear equations $AX = b$, then cv is a solution of system of linear equations $cAX = cb$, where $c \neq 0$.

Let $AX = b$ be a system of linear equations. If A is invertible, then $adj(A)x = b$ also has a solution.

Suppose C is the cofactor matrix of $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$. Let C_{ij} be the ij -th element of the cofactor matrix C . Answer the questions 5,6 and 7.

Find the value of C_{12} .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

Find the value of C_{22} .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -3

1 point

Find the sum of the elements in the third row of C .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

Level 2:

1 point

Which of the following square matrices of order 3 are the same as their adjoint matrices?

- ☐ Identity matrix.
- ☐ Zero matrix.
- ☐ Any scalar matrix.

☐ Any diagonal matrix.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Identity matrix.

Zero matrix.

1 point

Choose the set of correct options.

[Hint: For the first option, find out the adjoint matrix of an arbitrary square upper triangular matrix of order 22, i.e., take the matrix $\begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$ and find out its adjoint matrix.]

☐

The adjoint of a $2 \times 22 \times 2$ real upper triangular matrix is an upper triangular matrix.

☐

There exists a square matrix of order 33, such that $A = A^{-1} = \text{adj}(A)A = A^{-1} = \text{adj}(A)$.

☐

If $A = A^{-1}A = A^{-1}$, then $\det(A)\det(A)$ must be 11.

☐

If $A = A^{-1}A = A^{-1}$, then AA must be an identity matrix.

☐

If $A = A^{-1}A = A^{-1}$, then $A^2 = IA2 = I$.

☐

If $A^{-1} = \text{adj}(A)A^{-1} = \text{adj}(A)$, then $\det(A)\det(A)$ must be 11.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The adjoint of a $2 \times 22 \times 2$ real upper triangular matrix is an upper triangular matrix.

There exists a square matrix of order 33, such that $A = A^{-1} = \text{adj}(A)A = A^{-1} = \text{adj}(A)$.

If $A = A^{-1}A = A^{-1}$, then $A^2 = IA2 = I$.

If $A^{-1} = \text{adj}(A)A^{-1} = \text{adj}(A)$, then $\det(A)\det(A)$ must be 11.

1 point

If A and BA and B are squares matrices of order 22, then choose the set of correct options.

[Hint: Find out the adjoint matrix of an arbitrary square matrix of order 22, i.e., take the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and find out its adjoint matrix.]

☐

$\text{adj}(AB) = \text{adj}(A)\text{adj}(B)\text{adj}(AB) = \text{adj}(A)\text{adj}(B)$.

☐

$\text{adj}(AB) = \text{adj}(B)\text{adj}(A)\text{adj}(AB) = \text{adj}(B)\text{adj}(A)$.

☐

$\text{adj}(A + B) = \text{adj}(A) + \text{adj}(B)\text{adj}(A + B) = \text{adj}(A) + \text{adj}(B)$.

☐

$\text{adj}(A^T) = \text{adj}(A)^T \text{adj}(AT) = \text{adj}(A)^T$.

☐

$\text{adj}(A^{-1}) = \text{adj}(A)^{-1} \text{adj}(A^{-1}) = \text{adj}(A)^{-1}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\text{adj}(AB) = \text{adj}(B)\text{adj}(A)\text{adj}(AB) = \text{adj}(B)\text{adj}(A)$.

$\text{adj}(A + B) = \text{adj}(A) + \text{adj}(B)\text{adj}(A + B) = \text{adj}(A) + \text{adj}(B)$.

$$\text{adj}(A^T) = \text{adj}(A)^T \quad \text{adj}(AT) = \text{adj}(A)^T.$$

$$\text{adj}(A^{-1}) = \text{adj}(A)^{-1} \quad \text{adj}(A^{-1}) = \text{adj}(A)^{-1}.$$

Check Answers

Your score is: 0/10